Sec 6.6 (Application of Integration)

Work: The energy required to move an object over a distance. The units are Joules = Newtons xMeters

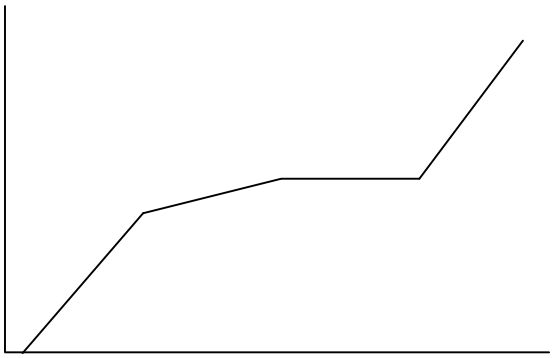
Under a constant force Work = Force x Distance Force is given by Newton's second law F=ma where m is the mass and a the acceleration.

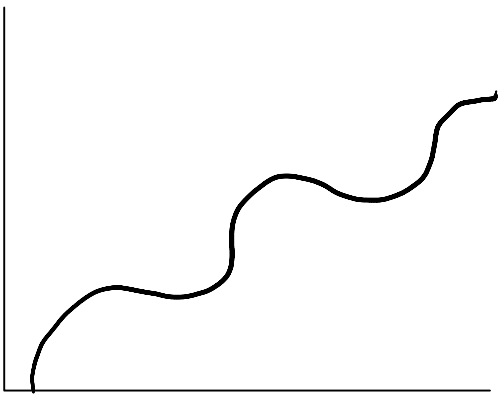
With a constant acceleration -> constant force work is defined as W=Fd

Example: Calculate the force required to lift a 20kg bag of cement 1m from the ground to the back of a pick up truck.

Sec 6.6 (Application of Integration)

What if we don't have a constant force?





Sec 6.6 (Application of Integration)

Work done from moving an object from a to b:

$$W = \int_a^b f(x)dx$$

Where f(x) is the force

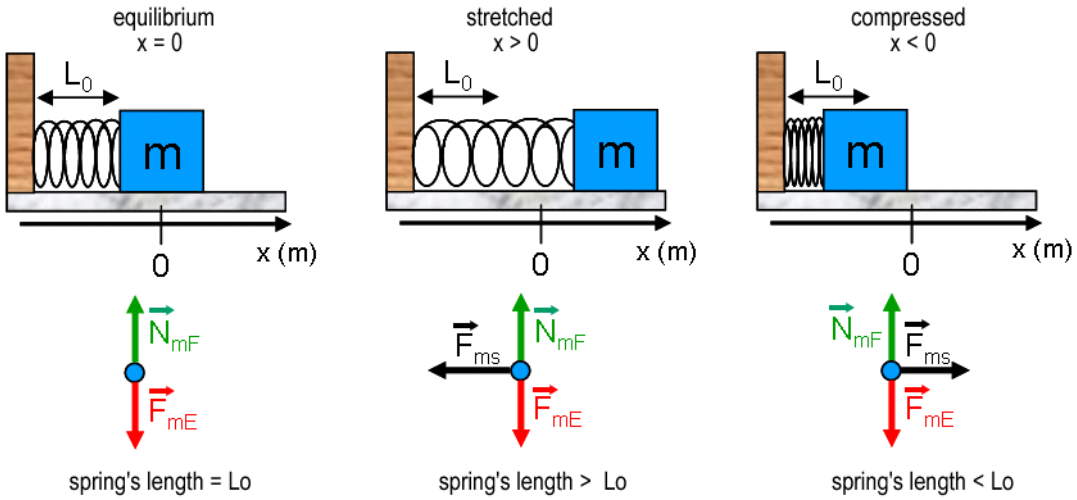
A particle is located a distance x meters from the origin, a force of f(x)=$\cos\left(\frac{\pi x}{3}\right)$ acts on it. How much work is done in moving the particle from x=1 to x=2.

Sec 6.6 (Application of Integration)

A heavy rope 50 ft long weighs .5 lb/ft and hangs over the edge of a building 120 ft high. How much work is done pulling the rope to the top of the building?

Sec 6.6 (Application of Integration)

Hooke's law: The restoring force of a spring is given by f=-kx where k is the spring constant (a property of the spring) and x the distance it is stretched beyond its natural equilibrium.

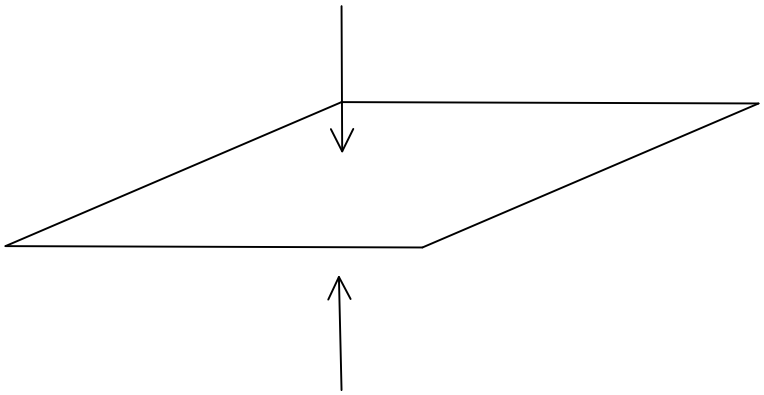


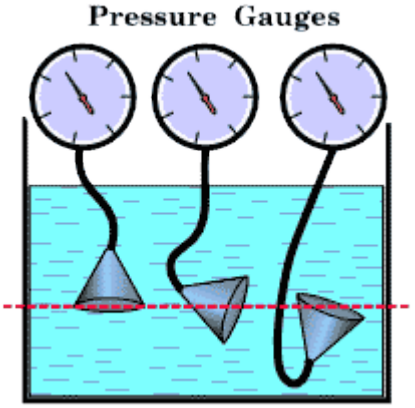
Example Calculate the work required to stretch a spring with spring constant k=4 a distance of 10 cm from equilibrium.

Sec 6.6 (Application of Integration)

Hydro static pressure.

F=mg P=F/A



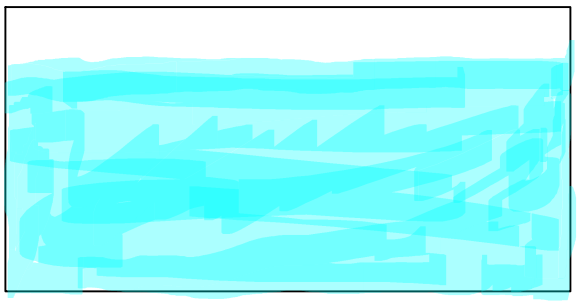


If we use 1 square meter for our area then the hydrostatic pressure at a depth d is $P = \rho g d$

Sec 6.6 (Application of Integration)

$P = \rho g d \quad F = PA$

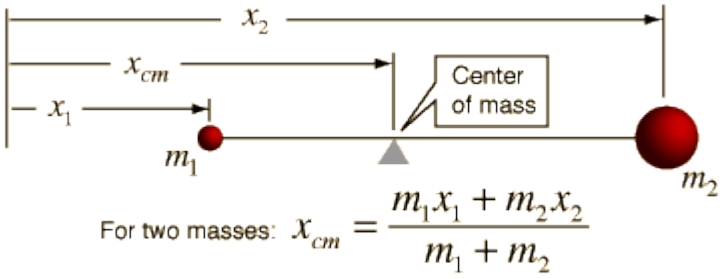
Suppose we have a rectangular dam which is 40 m wide and 20 m high is filled to the point where the water is 5 m from the top. Find the hydrostatic force on the dam. Is there a design that might be better then a rectangle?



Sec 6.6 (Application of Integration)

Moments and centers of mass

$$x_{cm} = \frac{\sum m_i x_i}{\sum m_i}$$



For two masses: $x_{cm} = \dfrac{m_1 x_1 + m_2 x_2}{m_1 + m_2}$

For two masses on a "sea saw" They balance when $m_1 x_1 = m_2 x_2$
That is when the moments are equal.

Center of mass in a plane:

$$x_{cm} = \frac{\sum m_i x_i}{\sum m_i}$$

$$y_{cm} = \frac{\sum m_i y_i}{\sum m_i}$$

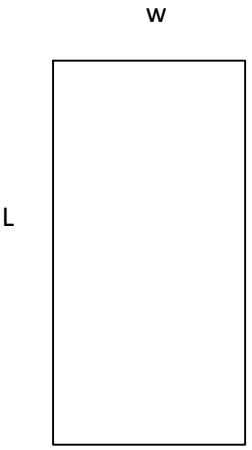
Moments about an axis

$$M_y = \sum m_i x_i$$

$$M_x = \sum m_i y_i$$

Sec 6.6 (Application of Integration)

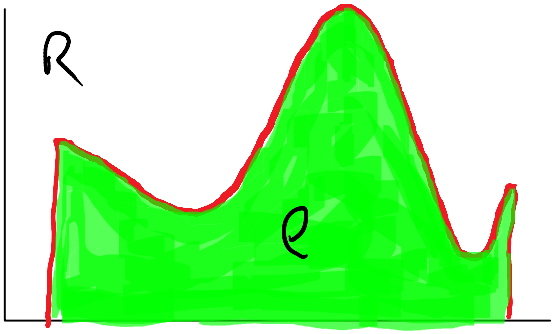
Centroid: The center of mass of a plane or 2d object.

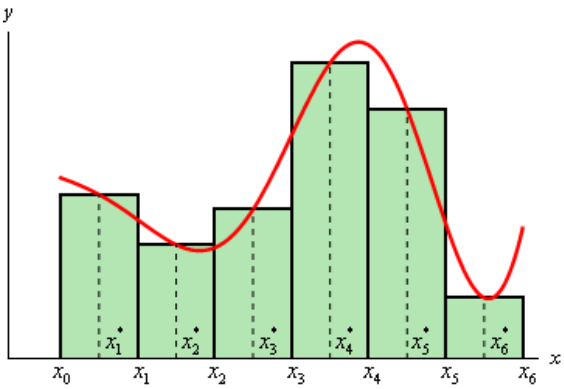


Until you learn about integrals over a plane we will need to use some physical principles to find the Centroid of a 2d region R. They are:

1.The symmetry principle says that if R is symmetric about a line l , then the centroid of R lies on l.
2.If R is reflected about the line l then R remains the came so its centroid remains fixed.

$$m = \rho A \ , x_{cm} = \frac{w}{2} \ , y_{cm} = \frac{L}{2}$$





Sec 6.6 (Application of Integration)

$M_y = \rho \int_a^b x f(x) dx$ $M_x = \rho \int_a^b \frac{1}{2} \big(f(x)\big)^2 dx$ $m = \rho \int_a^b f(x) dx$

Sec 6.6 (Application of Integration)

Find the Centroid of the region bounded by the curves $y = 4 - x^2 \ \ and \ \ y = 0$